Sai Preetam Umesh Sasankota

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.15
JSS Public School	Class XII (CBSE)	2023	94.20%
JSS Public School	Class X (CBSE)	2021	95.80%

WORK EXPERIENCE

Goldman Sachs

Software Engineering Intern -- Asset & Wealth Management

- Worked in the **Asset Transfers** engineering pod for **Private Wealth**, supporting high-value client transfers of **$1M+** and enabling **check and wire** functionality, where accuracy and reliability are critical for high-net-worth clients.
- **Optimized 59.3% of core payment & transfer user stories** by refactoring **26** duplicate end-to-end flows, replacing **12** flaky Selenium UI journeys with direct API validation, and automating **10** manual test paths.
- **Engineered a custom Model Context Protocol (MCP) server** integrating with the internal QA portal, enabling autonomous AI agents to fetch live batch execution telemetry and parse structured logs on demand.
- **Designed a deterministic Retrieval-Augmented Generation (RAG) pipeline** combining regex pattern matching with domain-grounded LLM inference, eliminating context noise and generating grounded fix proposals.
- **Reduced regression execution duration by 20%** across a 4-person QA engineering pod and deployed the internal solution to Cloud Managed Site Hosting.

PROJECTS

Nanosecond-Scale Matching Engine | *C++17, Zero-Allocation, CMake*

- Engineered a price-time priority matching engine achieving **10.4M+ messages/sec** with a **~48 ns median** and **~274 ns p99** latency, validated via RDTSC-based benchmarking across 1M-order simulations.
- Eliminated all dynamic allocation on the hot path by replacing deque/unordered_map with a pre-allocated memory pool, intrusive linked lists, and flat array-indexed order lookups, cutting p99 tail latency by **~98%** (14.4 µs -> 274 ns) and lifting throughput **~6.9x** (1.5M -> 10.4M msg/sec) over the baseline.
- Diagnosed and resolved a cache-locality bottleneck from pointer-chasing hash lookups on order cancel/modify, and measured ~20 ns of std::chrono per-call overhead, switching to RDTSC for accurate sub-100 ns profiling.

RISC-V 32-bit 5-Stage Pipelined Processor | *Verilog, Python*

- Implemented the microarchitecture of a **5-stage pipelined processor** in Verilog for the **RV32I ISA**, covering fetch, decode, execute, memory, and write-back stages.
- Optimized performance with **hazard resolution** techniques including **stalling, flushing, and data forwarding**.
- Verified functionality using a **Python-based assembler** to feed instructions into internal memory.

Efficient LLM Fine-Tuning & Serving Pipeline | *PyTorch, Hugging Face, LoRA, AWQ, vLLM*

- Fine-tuned a **Qwen2.5-1.5B model** for text-to-SQL translation using LoRA (updating only **0.6%** of parameters), which boosted exact-match query accuracy by **28%** (from 49% baseline to 77%) across 8,000 training examples.
- Optimized **model memory footprint** by quantizing the fine-tuned weights to 4-bit precision via **AutoAWQ**, achieving a **68%** reduction in VRAM usage (3.1GB down to 1.0GB) with less than a 2% degradation in task accuracy.
- Architected a **high-throughput inference API** with vLLM and FastAPI, achieving a **40x throughput improvement** (10 to 404 tokens/sec) over a naive Hugging Face baseline while maintaining 1.2s median latencies at 32 concurrent requests.

SCHOLASTIC ACHIEVEMENTS

NTSE Scholar - Rank **12** in state and top **800** out of **800,000+** students; awarded scholarship from **Government of India**

KVPY Scholar - Rank **833** out of **500,000+** students; awarded annual scholarship from **Indian Institute of Science (IISc)**

International Rank 8 in **SOF IMO** conducted by Science Olympiad Foundation

Codeforces Specialist (Max Rating: 1460), showcasing strong algorithmic and problem-solving skills.

Qualified IOQM (Indian Olympiad Qualifier in Mathematics) - Received a **Certificate of Merit** for being **top 3%** in the country

Invited to **IISc Bangalore** for an exclusive **5-day** academic program, selected from top-performing students nationwide.

SKILLS

Languages: C, C++, Python, JavaScript, SQL

Technologies: FastAPI, REST APIs, PostgreSQL, Redis, Docker, Git, Linux

AI/Machine Learning: PyTorch, Transformers, RAG, LLMs, MCP

Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, System Design, Distributed Systems

POSITIONS OF RESPONSIBILITY

Head Placement Coordinator -- Electrical Engineering Department

- Led over **150 students** in the department-wide placement process for the largest academic department at IIT Hyderabad.

Overall Head, Blockspace (Blockchain club):

- Headed the campus blockchain club, organizing hackathons, workshops, and speaker sessions to drive Web3 engagement.

Inter IIT Team Lead: Blockchain Problem Statement, Inter IIT Tech Meet

PR Assistant Manager, E-Cell : The Entrepreneurship club of IITH

Core Member: Electronika (Electronics club) and Glitch (Gaming club)